

02

PLAN

SPECIFIC

INTERVENTION

CENTERS

HEALTH SOCIOLOGY



**Comunidad
de Madrid**

Extreme heat is dangerous for everyone, ESPECIALLY FOR PEOPLE ELDERLY PEOPLE, AND ESPECIALLY THOSE LIVING IN RESIDENCES FOR THE ELDERLY.

During a heat wave, when temperatures remain abnormally high for
If the immune system fails after a couple of days, the defense mechanisms of older people can fail, causing serious health problems and even death.

In August 2003, the various heat waves left a number of deaths and a social collapse of magnitudes never experienced by Europeans in hundreds of years.

While it is difficult to make a rigorous count of deaths given that many countries do not keep a central record on the health impact of these phenomena, in the places where a count could be made, it was established that there were 25% more deaths than on the same dates in the summer of the previous year.

Some studies claim that the final death toll across Europe as a result of the high temperatures was 70,000, although others lower the figure considerably to 35,090; however, there is agreement that the greatest increase in the risk of death occurred among residents of care homes.

In June 2022, Spain experienced an episode of abnormally high temperatures.
It was the warmest for that date on record. Rising temperatures will make heat waves more intense, longer lasting, more frequent, and earlier in the year.

It is especially important that workers in residential care homes for the elderly and the directors of these centers, where the people most at risk of suffering the effects of a heat wave are located, are familiar with this document.

They are strongly recommended
that they take the measures recommended in this document before a heat wave occurs. The effects of heat occur rapidly, and for the measures to be effective, they must be taken beforehand.

Preparatory exams should be taken before the start of the summer.

WHAT ARE THE RISKS?
THE EFFECTS OF HEAT ON
HEALTH

The body normally cools itself through four mechanisms:

- radiation in the form of infrared rays;
- convection through water or air passing through the skin;
- conduction through a cooler object in contact with the skin;
- evaporation of sweat.

When the ambient temperature is higher than the skin temperature, the only effective mechanism for heat loss is sweating. Therefore, any factor that reduces the effectiveness of sweating, such as dehydration, lack of ventilation, tight clothing, or certain medications, can cause the body to overheat. Additionally, thermoregulation, which is controlled by the hypothalamus, can be impaired in the elderly and those with chronic illnesses, and potentially in those taking certain medications, making the body more vulnerable to overheating. Older women appear to be more vulnerable to the effects of heat than men, possibly due to having fewer sweat glands and being more likely to live alone.

This document describes the effects of overheating the body, which in the form of heat stroke, can be fatal.

However, the main causes of illness and death during a heat wave are respiratory and cardiovascular diseases.

Several studies in the UK demonstrate a linear relationship between temperature and weekly mortality, with an estimated 75 additional deaths per week for every degree Celsius increase in temperature. Another key factor is



the effect of heat on the cardiovascular system.

To stay cool, large amounts of extra blood need to circulate to the skin.

This puts the heart under strain, which for older people and those with chronic health problems can be enough to trigger a heart problem.

Sweating and dehydration affect electrolyte balance.

For people taking medications that affect electrolyte balance or heart function, this can also be a risk. Medications that affect the ability to sweat, thermoregulation, or electrolyte balance can make a person more vulnerable to the effects of heat.

These medications include anticholinergics, vasoconstrictors, antihistamines, medications that affect kidney function, diuretics, psychoactive medications, and antihypertensives.

The relationship between the

High ambient temperatures and associated dehydration increase the risk of infections caused by Gram-negative bacteria, particularly *Escherichia coli*. The risk is even higher for people over 65, so it is crucial to emphasize the importance of ensuring adequate fluid intake in older adults during periods of high temperatures to reduce the risk of infection.

Whatever the underlying cause of heat-related symptoms, the treatment is always the same: move the person to a cooler place and cool them down.

RELATED DISEASES WITH THE HEAT

- Increased risk of cardiovascular and respiratory diseases - which cause most of the excess mortality during a heat wave.
- Heat cramps - caused by the dehydration and electrolyte loss, often after exercise.
- Heat rash - small, red, itchy bumps.

- Heat edema - mainly in the ankles, due to vasodilation and fluid retention.
- Heat syncope - dizziness and fainting, due to dehydration, vasodilation, cardiovascular diseases, and certain medications.
- Heat exhaustion – is more common. It occurs as a result of water or sodium loss, with nonspecific features of malaise, vomiting, and circulatory collapse, and appears when the temperature is between 37°C and 40°C. If left untreated, heat exhaustion can develop into heat stroke.
- Heat stroke - can develop into a
Heat stroke is the point of no return at which the body's thermoregulation mechanism fails. This leads to a medical emergency, with symptoms including confusion, disorientation, seizures, loss of consciousness, hot and dry skin, and a body temperature above 40°C for a period of 45 minutes to eight hours. It can cause cell death, organ failure, brain damage, or death. Heat stroke can be classic or exertional (for example, in athletes).

ADVANCE PREPARATIONS

It is recommended to create a checklist based on the following actions, which should be completed every year before the start of the summer.

Long-term planning (all year round)

Work with your healthcare system contacts (local public health, primary care and geriatric liaison at your local hospital) to develop long-term plans for heat wave preparedness.

Make environmental improvements to provide a safe environment for your users in the event of a heat wave.

Prepare your center's business continuity plans to cover the eventuality of a heat wave

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(e.g., medicine storage, IT resilience, etc.)

Work with your staff to raise awareness about the impact of a severe heat wave and the importance of reducing risks.

The building and its surroundings

Check that the windows can be covered, preferably with light-colored, reflective curtains rather than metal blinds and dark curtains, which can worsen the situation. If these are already installed, check that they can be raised.

Check that there are no problems opening windows while respecting safety measures.

Increase exterior shade, in the form of shutters or other elements that provide shade, trees or leafy plants.

Reflective paint can also help keep the building cool. Increasing exterior vegetation, especially in concrete areas, is beneficial as it increases moisture content and acts as a natural air conditioner.

Insulating walls and partitions helps keep the building warm in winter and cool in summer. Contact your community's energy efficiency officer or your electricity company to see what grants are available.

Create cool rooms or zones. High-risk groups who are vulnerable to the effects of heat are physiologically unable to cool themselves efficiently when temperatures rise above 26°C. Therefore, all care homes for the elderly should be able to provide a room or zone that maintains a temperature of 26°C or lower.

Cool areas can be created by providing adequate indoor and outdoor shading, ventilation, the use of indoor and outdoor plants, and, if necessary, air conditioning.

Make sure staff know which rooms are the easiest to maintain

fresh ones and which ones are the most difficult and review the distribution of residents according to those who are at greater risk.

Indoor thermometers should be installed in each room where vulnerable individuals spend a lot of time (bedrooms, living areas and dining rooms) and, during a heat wave, indoor temperatures should be checked at least four times a day.

Electric fans can provide relief if temperatures are below 35°C (note: using fans in temperatures above 35°C does not prevent heat-related illnesses. Furthermore, fans can cause excessive dehydration. It is recommended to position the fan

Keep a safe distance from other people, do not point it directly at the body, and drink fluids regularly – which is especially important for bedridden patients.

bed).

RISK LEVELS

The following levels refer to the levels of the main plan to prevent heat waves, based on the three risk levels established according to the maximum temperatures forecast by the State Meteorological Agency (AEMET). They are visualized with traffic light colors and are: normal (green), caution (yellow), and high risk (red).

GREEN LEVEL

Normal: Expected temperatures not exceeding 36.5°C: summer heatwave preparedness program – advance preparations

Work organization

- Make sure the plans continuity of the center are implemented correctly (it is necessary to



have sufficient staff available to implement the necessary measures in the event of a heat wave).

- Get additional help from residents' family members and volunteers.
- Identify the cool areas in the center.
- Increase staff awareness of heat-related illnesses and protective health measures.

Facilities

- Check that you have an adequate supply of fans and sprinklers water.
- Verify that water and ice are widely available.
Make sure you have a supply of electrolyte solutions.
Oral rehydration, orange juice and bananas to help maintain electrolyte balance in people taking diuretics.
- Make sure the cold drinks are distributed regularly in case of a heat wave.
- Plan to adapt menus to cold meals (preferably with a high water content, such as fruit and salads) in consultation with residents.

Residents

- Make sure you know who is at greater risk (see Additional Notes on page 14) - ask your primary care staff (or referring geriatrician) if you are unsure.
- Make sure you have protocols in place for monitor residents in greater

risk and to provide additional care and support (it will be necessary to monitor room temperature, body temperature, pulse rate, blood pressure and dehydration).

- Ask the doctors at the Primary Care Center
Primary care for at-risk residents about possible changes in treatment or medication in the event of a heat wave and review residents on multiple medications.

- Check that residents have access to light, loose-fitting cotton clothing.
Please note that mattress protectors and plastic mattresses can get very hot during heat waves.

YELLOW LEVEL

Caution. Type 1 alert: This level is activated when the maximum temperature forecast for that day or any of the following four days is above 36.5°C, without there being more than three consecutive days at this temperature.

- Identify high-risk residents.
- If indoor temperatures exceed 26°C, high-risk individuals should be moved to a cooler area at 26°C or lower – for those who cannot be moved or who might become too disoriented if moved.

Take steps to cool them down (e.g., fluids, cold compresses) and increase vigilance.

- Check the weather forecast on the radio, television, internet or press.

- Check that staff and others collaborators, such as volunteers, know what to do during a heat wave.
- Suggest that all residents consult their primary care physician about possible changes to their treatment and/or medication; consider prescribing rehydration salts oral for those taking high doses of diuretics.
- Check the temperatures
Indoor temperatures are regularly recorded during periods of higher temperatures in all areas where patients reside.
- Communicate the alerts to staff and
Make sure they are aware of the heatwave plans.
- Prepare cool areas and provide regularly wet towels and cold foot baths.
- Make sure you have enough staff.

RED LEVEL

High risk. Alert type 2: This level is activated when the maximum temperature forecast for that day or any of the following four days is above 38.5°C or when more than three consecutive days in this period are expected to have maximum temperatures above 36.5°C.

- Keep the center as cool as possible
- Make sure you have taken the steps indicated for Level 1 and 2.
 - Activate the plans to maintain the continuity of the center's activity – including a possible increase in demand for services.

- Increase outdoor shading. Watering the outdoor soil helps cool the air (avoid slip hazards and check local watering restrictions).
- of water during periods of drought before using the hoses).
- Keep curtains and windows closed while the outside temperature is higher than the inside temperature.
- When the outside temperature is around
If the indoor temperature drops below normal, open the windows. This may not happen until very late at night or in the early morning.
- Discourage physical activity for the residents and outings during the hottest hours of the day (from 11:00 to 15:00).
- Check the temperatures
Indoor temperatures are regularly recorded during periods of higher temperatures in all areas where patients reside.
- Make sure that staff can help and advise users.
- Take full advantage of cooler nighttime temperatures to cool the building with ventilation.
- Reduce the indoor temperature by turning off unnecessary lights and electrical appliances.
- Consider changing the visiting hours
in the mornings or late afternoons/evenings to avoid the heat during the afternoon due to the concentration of people.

- Monitor residents
- Monitor body temperature, heart and respiratory rate, blood pressure, and hydration levels.



- Be alert to any changes in behavior, especially excessive sleepiness.

- Be alert for signs of pain
headache, unusual tiredness, weakness,
dizziness, disorientation, or trouble sleeping.

Reducing the health risks of heat

- Encourage residents to stay in the cooler areas of the building as much as possible
as much as possible.

- Move residents so that each
Spend time in the cool room/area (below 26°C) -
give priority and extra time to high-risk residents or
those showing signs of distress (including
increased body temperature); for those who
cannot be moved or who might become too
disoriented when moved, take steps to cool them
down (e.g., fluids, cold compresses) and
increase monitoring.

- Monitor residents' fluid intake, providing them with
cool drinks regularly, especially if they cannot
always drink without assistance.

Remember the importance of increasing fluid intake during periods of high temperatures to reduce the risk of bloodstream infections caused by Gram-negative bacteria.

Oral rehydration salts are
Bananas, orange juice, and occasionally salty
snacks are recommended for those taking high
doses of diuretics. They can also help replace salts
lost through sweating.

- Advise residents to avoid caffeine (coffee, tea, cola drinks), very sweet drinks, and alcohol.

- Encourage residents to wear light, loose-fitting
cotton clothing to absorb sweat and
prevent skin irritation.

- Moje o rocíe regularly con
Apply cold water to exposed parts of the
body. A damp cloth on the back of the neck
helps regulate temperature.

- Provide cool showers or baths if possible.

EMERGENCY TREATMENT

If you suspect someone has suffered heatstroke, call 112.

While waiting for the ambulance:

- Take the person's temperature.
- If possible, move it to a cooler place.
- Cool her down as quickly as possible by giving her a cold shower, spraying her with water or wrapping her in a wet sheet, and using a fan to create an air current.
- Encourage her to drink fluids, if she is conscious.
- Do not give aspirin or acetaminophen.

ADDITIONAL NOTES

Risk groups include:

- Older people, especially those over 75, or those who live alone and are socially isolated, or in a care home

older people;
- People with chronic and serious illnesses, including heart conditions, diabetes, respiratory or kidney failure,

Parkinson's disease or mental illness
serious. Medications that potentially affect kidney function, sweating, thermoregulation, or electrolyte balance may make this group more vulnerable to the effects of heat;

- those who cannot adapt their behavior to stay cool, including those with Alzheimer's or a disability, or who are bedridden.

You can choose between receiving information via email or SMS messages on your mobile phone, or both at the same time.

It is important to note that subscription to this type of service is only valid during the campaign in course.

General information about heat and health

Further information on heat and health can be found on the website <https://www.comunidad.madrid/servicios/salud/calor-salud>.

MORE INFORMATION

Daily Heat Wave Information Bulletin

Until September 15th, it is possible to access the Daily Heat Wave Information Bulletin (<https://www.comunidad.madrid/servicios/salud/boletininformacion-olas-calor>) where those responsible for the centers can access information on the evolution of the maximum temperatures expected for the day of the consultation and the next 4 days.

Information on alert levels

The Madrid Regional Ministry of Health has action plans in place to address extreme temperatures: the Heatwave Effects Monitoring and Control Plan and the Cold Effects Monitoring and Control Plan. The main objective is to minimize the health risks and effects of very high or very low temperatures recorded during the summer and winter periods, respectively.

It is recommended that directors and healthcare staff of the centers register (from the following link: https://gestion3.madrid.org/soca_web_inter/Ciudadano.icm) in the information subscription service for possible Heat Waves and Cold Waves depending on the time of year.

The Heatwave Monitoring and Control Plan is in effect until September 15th, and the Cold Weather Monitoring and Control Plan is in effect from December 1st to March 31st.